

SQL Sales Analysis Report

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This project focused on performing data cleaning, exploration, and business analysis on a sales transaction dataset using SQL for end-to-end analysis. The objective of the project was to develop SQL querying skills for extracting, transforming, and analyzing structured business data to generate meaningful insights.

The project began with data familiarization, where the dataset structure, column names, and data types were inspected using SQL queries such as `SELECT`, `COUNT`, and `DESCRIBE`. During this phase, an encoding issue was identified in the `ORDERNUMBER` column and corrected by renaming the column to improve consistency and query readability. The dataset contained a total of 2823 records and included transaction-related fields such as customer information, product categories, geographic details, sales values, and order status.

Data cleaning procedures were then carried out to improve data quality and analytical readiness. Missing value checks were performed separately for numerical and text columns. Missing or blank values were identified in `STATE`, `ADDRESSLINE2`, and `POSTALCODE`. Since these fields were important for geographic analysis, the missing entries were standardized using a placeholder value, "Unknown", to maintain grouping consistency during analysis and visualization. Duplicate record checks were also conducted to ensure data integrity, and additional validation checks were performed to identify zero or negative sales values and check for potential outliers in transaction amounts.

After data preparation, exploratory, and business-focused SQL analysis was performed using aggregation functions, grouping operations, conditional logic, subqueries, and window functions. Key analyses included total revenue calculations, revenue trends over time, top-performing product lines, customer segmentation, geographic sales distribution, and deal size contribution analysis. Advanced SQL concepts such as `CASE`, `HAVING`, `RANK`, `DENSE_RANK`, `ROW_NUMBER`, and `LAG` were used to generate deeper business insights and identify high-performing customers and products.

This project was completed using MySQL Workbench and Power BI.